FACULTY OF COMPUTING
SESSION 2025/2026 SEMESTER 2

SECP2613-02 SYSTEM ANALYSIS AND DESIGN (WBL)

PROJECT 1 PHASE 1

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1.0 INTRODUCTION

It cannot be understated how vital early diagnosis is when it comes to providing timely assistance for children with autism. However, even though this aspect is extremely important, it should not be overlooked that many of the approaches that are currently being utilized in relation to the problem tend to be scattered, expensive, and unavailable both for families and healthcare workers alike.

The suggested idea for the project revolves around developing a prototype called "Autism Early Screening Support System". The purpose of creating such a system is to provide a technological means that could facilitate the screening process, make it easily accessible, and offer accurate information. In order to make sure that the system becomes widely available to potential users, it will be developed for both the mobile platform and the web.

The system will benefit three main stakeholder groups, which include parents, autistic children, and a non-profit organization (NGO). As for the system owner, the NGO will own the system. The core functionality of the system will involve the incorporation of different modules, such as M-CHAT-R (Modified Checklist for Autism in Toddlers, Revised) autism screening module and profiling according to the BPPOKU Pindaan 2019 (Bahagian A). This system will seek to solve the present challenges in data management, which are impeding efficient and systematic provision of services by both the health care and education sectors.

2.0 BACKGROUND STUDY

Organizational Background Study

The system owner is an NGO which supports families and children affected by autism. It will serve as the central point that would help to solve the problem of the scattered and very resource-intensive autism screening currently faced today. As an NGO that owns this system, there is a desire to bring together both health care and education systems in order to solve problems of poor awareness and the absence of standardized measures.

3.0 PROBLEM STATEMENT

The efficient support of autistic children becomes possible only due to the timely screening in the early stages of life. The main issues that arise when dealing with autism screening processes include the following:

3.1. The Disjointed and Unavailable Nature of the Process

Current methods of assessing whether a kid is developing normally require excessive resources and accessibility problems. Moreover, the fragmented nature of the process may make it difficult for parents to find out what to do first when suspecting that their child is underdeveloped.

3.2 Overreliance on Traditional Screening Practices

Despite the availability of more advanced technologies, traditional screening procedures are still very dependent on manual labor. Due to this fact, as well as other reasons related to the lack of communication and cooperation, screening often becomes delayed.

3.3 Existence of Systemic Obstacles and Absence of Integration

In most cases, there is a low degree of awareness as well as the absence of standardization. In addition, the lack of integration between the health care and education sectors leads to a "silo mentality" situation when the exchange of information by the parties accountable for a child’s development is rather poor.

3.4 No Technology-Based Centralization Solution

The lack of centralization, with no technology-based solution, leaves room for inefficiencies in terms of data management as well as coordination of activities by various NGOs and parents. Tracking of results for screening tests, such as M-CHAT-R, and managing profiles based on BPPOKU guidelines will not be efficient without technological solutions.

As a result of such issues, the problem area described above is one wherein a delay in providing support for children with symptoms of autism may occur.

4.0 PROPOSED SOLUTIONS

The recommended solution is known as the Autism Early Screening Support System, which is an advanced two-fold application Web and Mobile technology system. The system combines standard medical screening procedures with daily management systems to create an efficient technological platform for the use of parents, children, and NGOs.

4.1 Core System Infrastructure and Security

The basis of the system lies in data security and privacy.

4.1.1 Authentication and Access Controls

The system uses safe methods for logging into and out of the system, which ensures that confidential data is protected by not allowing access to people who are not authorized to see the data.

4.1.2 Registration and Account Management

A separate module of the system is used to manage users' accounts.

4.1.3 Data Security and Privacy

Considering the confidentiality of data in question, data security is a critical feature of the core infrastructure of the proposed solution, as it will serve as a reliable environment for the formative years intervention program.

4.2 Standardized Screening & Profile Management

This module digitalizes the old screening processes manually performed to guarantee accuracy and fast intervention.

4.2.1 Digital Profile Management (BPPOKU Pindaan 2019)

The program will digitalize Bahagian A (Biodata) from the BPPOKU (Pindaan 2019) form. This process will include the following important pieces of information:

- ➢ Personal Details

  Name, MyKad / MyKid identification, and contact details.

- ➢ Education & Professional Experience

  Education level and parents/guardians' employment.

- ➢ Details of Next of Kin

  The system collects detailed information about relatives close to the person for further communication in case of necessity.

4.2.2 M-CHAT-R Screening Integration

This feature includes a test with 20 questions performed according to the M-CHAT-R standard. The users provide "Yes" or "No" answers regarding children's behavior, such as do they react to a gesture or respond when called?

4.2.3 Automated Scoring Algorithm of Risk Level

The system will be able to provide risk levels automatically, making sure that each child undergoes an evaluation process that is not only consistent but also standardized and objective in nature. For those children with results falling within the Low Risk category, which would include any score of 0-2, there would not be a need to pursue any additional actions unless there are any future worries. Those with scores ranging from 3-7 are considered to have Moderate Risk, automatically prompting notification about the requirement of taking some actions to further examine their behavior. In instances where the scores range from 8-20, the result will be classified as a High Risk automatically, skipping other steps to indicate the urgent requirement of referral of the client for professional diagnostic testing.

4.3 Support, Analytics and Specialized Management

In addition to the diagnosis, this system features modules which enable monitoring and assistance on a continuous basis.

4.3.1 Automated Notification System

To resolve the issue of "delayed referrals" in the conventional approach, this module helps generate automated alerts regarding upcoming screenings or follow-up consultations.

4.3.2 Reporting and Analytics

Data analytics provide an opportunity to understand how the development of a child evolves throughout time, as visualized by parents or NGOs.

4.3.3 Behavior Monitoring

This module monitors the emotional state and mood of the child, enabling caregivers to identify any possible triggers for certain behaviors. These are some of the functions of this module:

- Behavior Log Processor

  A real-time function that enables the caregiver to log emotional status, particular behavioral instances, and the corresponding environmental antecedents.

- Trigger Analyzer

  A pattern-matching algorithm that scans historical logs to identify correlations between environmental factors that repeat themselves and behavioral disruptions that occur again.

- Mood Trend Evaluator

  A computation function that calculates and charts baseline emotional movements over specified timelines, providing parents and NGOs with empirical evidence of structural improvement.

4.4 Feasibility Study

4.4.1 Technical Feasibility

The technical feasibility of the Autism Early Screening Support System is possible considering the team's strengths in data engineering and structured programming. The front end can be developed with HTML5 to maintain a clean format that will allow the M-CHAT-R assessment procedure to proceed efficiently. For the back end, C++ will be used together with OOP methods to provide modularity in processing data using encapsulated classes for user accounts and test results. Data quality will be ensured using SQL Relational Database, especially in protecting confidential BPPOKU (Pindaan 2019) biodata and monitoring the individual's developmental history. Project management will be implemented systematically using the Github platform for versioning and Kanban charts for visualizing tasks.

4.4.2 Operational Feasibility

Operatively, the aim is to address the existing problem of fragmentation prevalent in the manual autism screening method. By adopting the BPPOKU standard (Pindaan 2019) into a technological medium, the proposed system will conform to existing documentation practices in Malaysia's welfare and healthcare institutions, thus facilitating easier implementation by NGOs. The system is tailor-made for the unique needs of three key players: helping parents get access to a screening tool, providing children with assistance through Routine and Behavior Management, and offering NGOs an online platform for analytics. In addition to doing away with the delays associated with manual paper work, automation guarantees that the screening process becomes a dynamic and ongoing exercise for children.

4.4.3 Economical Feasibility

Economic viability of the Autism Early Screening Support System is determined through quantitative assessment of costs involved in its developmental stage against potential savings in operations. Through computerization of the M-CHAT-R test results and BPPOKU data handling process, the proposed system will create great value through cost reduction and minimizing human error. Based on the calculation below, the project is regarded as an extremely viable venture due to the high Profitability Index (PI) of 1.96, which is very much higher than the threshold value of 1.0. It implies that for every dollar invested in the system, there will be a two-fold return.

Estimated Costs

Development Cost Category	Estimated Value
System Analysis and Design	RM 850
Frontend Development (HTML)	RM 1200
M-CHAT-R Algorithm Development (C++/OOP)	RM 1100
Database & Data Structure Setup (SQL)	RM 850
Testing and Debugging	RM 600
Documentation & User Guide	RM 500

Production Cost (Annual)	Estimated Value
Cloud Database Hosting (SQL)	RM 650/year
Security & Data Encryption	RM 350/year
System Maintenance & Debugging	RM 900/year
Notification API	RM 300/year
Internet & Clous Utilities	RM 400/year

Estimated Benefits

Benefits Category	Estimated Value
Staff Productivity Gains	RM 4200/year
Reduced Physical Storage & Paper Costs	RM 1500/year
Avoidance of Manual Scoring Errors	RM 2800/year

Assumptions

Assumptions	Rate
Discount rate	8%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefits)	0.9
Annual change in production costs	8%

Annual change in benefits	5%

Cost-Benefit Analysis (CBA)

Costs	Year 0	Year 1	Year 2	Year 3	Year 4
Development Costs					
System Analysis and Design	935				
Frontend Development	1320				
M-CHAT-R Algorithm Development	1210				
Database & Data Structure Setup	935				
Testing and Debugging	660				
Documentation & User Guide	550				
Total	5610				
Production Costs					
Cloud Database Hosting		715	772	834	901
Security & Data Encryption		385	416	449	485
System Maintenance & Debugging		990	1069	1155	1247
Notification API		330	356	385	416
Internet & Clous Utilities		440	475	513	554
Annual Production Costs		2860	3088	3336	3603
Present Value (PV)		2648	2647	2648	2648
Accumulated Costs		8258	10905	13553	16201

Benefits	Year 0	Year 1	Year 2	Year 3	Year 4
Staff Productivity Gains		3780	3969	4167	4376
Reduced Physical Storage & Paper Costs		1350	1418	1488	1563
Avoidance of Manual Scoring Errors		2520	2646	2778	2917
Total		7650	8033	8433	8856
Present Value (PV)		7083	6887	6694	6509
Accumulated Benefits		7083	13970	20664	27173
Gain or Loss		-1175	3065	7111	10972
Profitability Index	10972 / 5610 = **1.96**				
Justification	Good investment because PI value is greater than 1.				

5.0 OBJECTIVES

The primary objective of this project is the development of a secure cloud-based Autism Early Screening Support System that leverages current screening methods and communications technologies to enhance the process of early diagnosis and the formation of continuous monitoring procedures. Traditional manual and paper-based practices will be replaced by the proposed system, which will act as an automated alternative to link parents with NGOs and promote effective cooperation for diagnostics and administration. Through this project, a contribution will be made toward achieving SDG 3 (Good Health and Well-being), as well as SDG 10 (Reduce Inequalities) relating to children suffering from autism, their parents, and NGOs.

In order to achieve the primary objective, the following secondary objectives will be met according to the standard SDLC phases:

➢ Gathering and analyzing requirements (Planning & Analysis Phase): In order to capture the essential functional requirements from the caregivers and NGO administration, it will be necessary to define the system specification needed in order to automate the process using the M-CHAT-R 20-question screening through a computerized method.

➢ Designing of the system architecture (Design Phase): It will be important to establish a database schema for cloud storage of all information related to clients' profiles based on the BPPOKU (Pindaan 2019) standards in Malaysia, along with user interface wireframes in order to ensure easy navigation.

➢ Development and Engineering of modules (Development Phase): To implement the programming of the basic core functionalities, particularly in the design of a Behavior Monitoring module that will have automated logic of monitoring emotions, analyzing mood trends, and detecting any triggers for behavior.

➢ Installation of components (Implementation Stage): To install the fully coded application by implementing a scalable database management tool and employing the GitHub configuration management approach for successful integration of the entire system.

➢ Testing module functions (Testing Stage): To conduct tests and verify the effectiveness of behavioral logging scripts, trigger detection, and reporting modules to ensure that full data accuracy is achieved and the whole system operates efficiently and seamlessly.

➢ Evaluating and maintaining the platform (Evaluation & Maintenance Stage): To evaluate the functionality of the final module in terms of meeting the baseline requirements of the NGO in relation to administrative efficiency and data security.

6.0 SCOPE OF THE PROJECT

The scope of the Autism Early Screening Support System covers the design and development of the centralized, dual-platform system that's accessible via both web browser and mobile interfaces.

6.1 Areas Covered By the System

The system is designed to provide a comprehensive digital access for early autism detection and management. The scopes of the system are defined below to align stakeholder expectations and clarify the purpose of the prototype.

6.1.1 What Will Be Included

System Components	Areas Covered By the System
Core Modules	➢ Secures authentication and access control ➢ User registration and account management ➢ BPPOKU (Pindaan 2019) Bahagian A digital profile forms
Screening Tools	➢ M-CHAT-R screening tool (20-question digital checklist) ➢ Automated risk scoring algorithm (Low/ Moderate/ High)
Support features	➢ Automated notification and reminder system ➢ Reporting and analytics dashboard
Specialized add-ons	➢ Daily Routine Management Module ➢ Behavior Monitoring Module

6.1.2 What Will Not Be Included

With respect to the system, formal clinical diagnosis by specialists are explicitly excluded by the system as the objective is primarily focused on early screening and does not replace professional clinical diagnostic assessments.

6.2 Project Deliverables

The following deliverables are to be produced as tangible outputs of the projects. These outputs are organized into three categories: core system components, screening and profile management, and add-on features.

6.2.1 Core System Components

➢ Web and mobile prototype

A fully functional frontend interface is to be developed where it is accessible on both web browsers and mobile devices to be used by parents and NGO staff.

- ➢ Secure authentication and access control

  A login and logout mechanism is to be implemented to prevent unauthorized access to confidential information.

- ➢ SQL Relational Database

  A structured database is to be established to store child profiles, screening histories, and developmental records securely.

- ➢ User registration and account management module:

  A dedicated module is to be developed to manage the accounts of parents, children, and NGO administrators.

6.2.2 Screening and Profile Management

- ➢ M-CHAT-R digital screening module

  A 20-question digital checklist is to be implemented, where Yes or No responses are recorded to assess toddler behavior in accordance with the M-CHAT-R standard.

- ➢ Automated risk scoring algorithm

  Results are to be automatically classified into three risk levels, which are Low (0-2), Moderate (3-7), and High (8-20) and with corresponding follow-up actions triggered without manual intervention.

- ➢ BPPOKU digital profile system

  Bahagian A forms capturing personal details (MyKard/MyKid), educational background, employment information, and next-of-kin data are to be digitalized in accordance with BPPOKU (Pindaan 2019) requirements.

6.2.3 Support and Add-On Features

- ➢ Automated notification system

  Automated alerts and reminders are to be generated for upcoming screenings and follow-up consultations to prevent delayed referrals.

- ➢ Reporting and analytics dashboard

Visual data insights related to a child's developmental progress over time are to be made accessible to parents and NGO administrators.

- Daily Routine Management module

  Visual schedules and structured daily routines are to be provided to support consistency in the activities of children with ASD.

- Behavior Monitoring module

  The emotional state and mood patterns of children are to be tracked, enabling caregivers to identify behavioral triggers and effective intervention strategies.

- Documentation and user guide

  A comprehensive user guide is to be prepared to facilitate the onboarding of NGO staff and parents onto the system.

7.0 PROJECT PLANNING

7.1 Human Resource

No	Role	Responsibility	Member
1.	Advisor	● Provide academic guidance and supervision throughout the project development process ● Monitor project progress and evaluate project quality ● Provide recommendations and feedback for system improvement	Dr. Muhammad Iqbal Tariq Bin Idris
2.	Project Manager	● Plan and manage the overall project schedule ● Act as the main bridge between stakeholders and the project team ● Organise and oversee the project team, including assigning tasks and responsibility to team members ● Manage GitHub repository	Qistina Batrisyia Binti Noor Mohd Azlan
3.	Front-end Developer	● Develop and design the user interface of the system ● Connecting the user interface to the backend APIs ● Converting UI / UX design prototypes into operational website	Yong See En
4.	UI / UX Designer	● Design wireframes, user flows and interactive prototypes ● Establishing visual design systems (color palettes, typography, UI components) ● Ensure consistency in interface design and navigation flow ● Testing designs with real users for feedback	Ng Xuan Yee

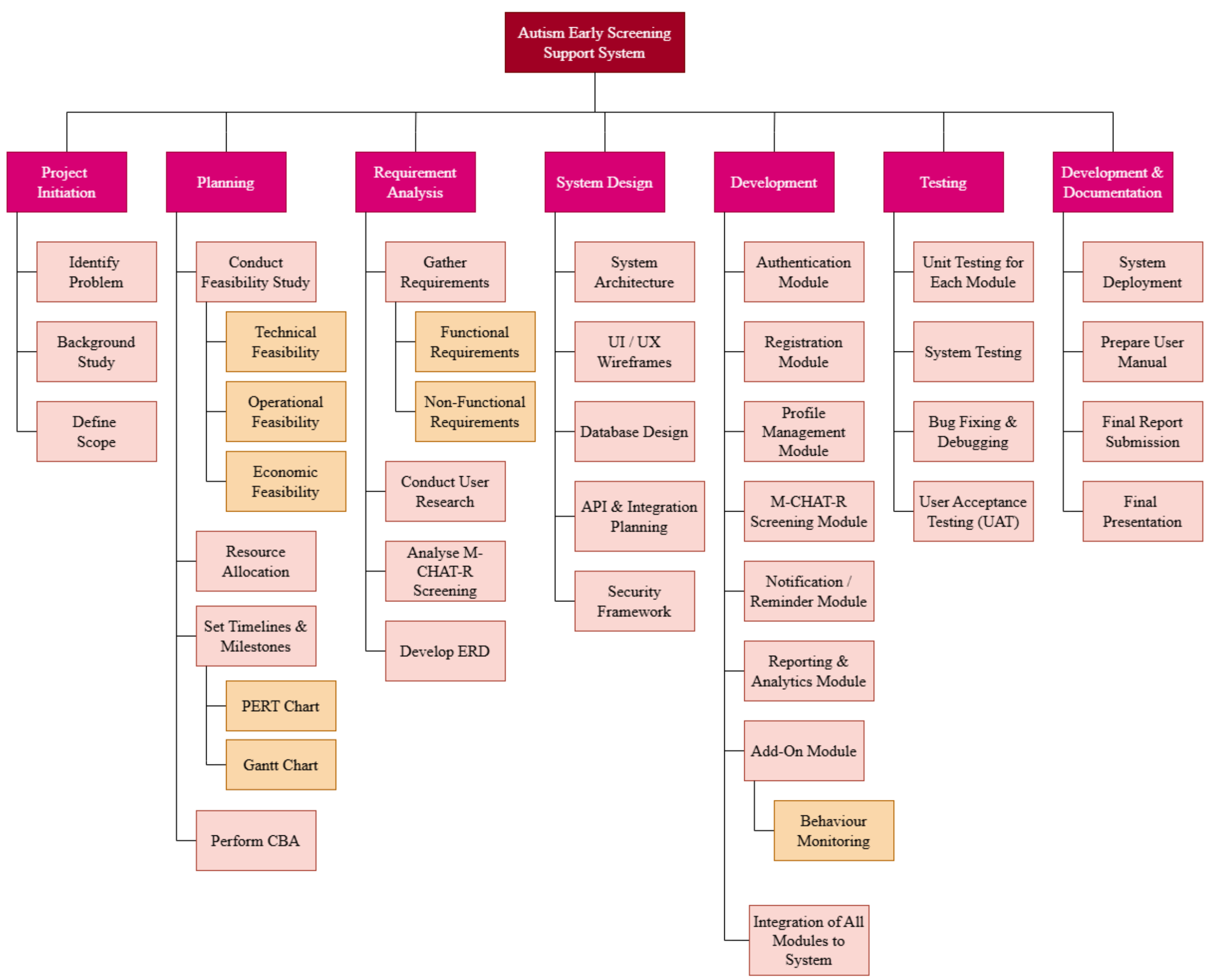
5.	Back-end Developer	- Designing and managing databases - Develop backend system functionalities and logic - Creating and maintaining APIs to manage information - Monitor and troubleshoot system bottlenecks	Cheng Zhi Min

7.2 Work Breakdown Structure (WBS)

1. Project Initiation
    - 1.1 Identify Problem
    - 1.2 Background Study
    - 1.3 Define Scope
2. Planning
    - 2.1 Conduct Feasibility Study
        - 2.1.1 Technical Feasibility
        - 2.1.2 Operational Feasibility
        - 2.1.3 Economic Feasibility
    - 2.2 Resource Allocation
    - 2.3 Set Timelines & Milestones
        - 2.3.1 PERT Chart
        - 2.3.2 Gantt Chart
    - 2.4 Perform Cost Benefit Analysis (CBA)
3. Requirement Analysis
    - 3.1 Gather Requirements
        - 3.1.1 Functional Requirements
        - 3.1.2 Non-Functional Requirements
    - 3.2 Conduct User Research
    - 3.3 Analyse M-CHART-R Screening
    - 3.4 Develop Entity Relationship Diagram (ERD)
4. System Design
    - 4.1 System Architecture
    - 4.2 UI / UX Wireframes
    - 4.3 Database Design
    - 4.4 API & Integration Planning
    - 4.5 Security Framework
5. Development
    - 5.1 Authentication Module (Login / Logout / Security)
    - 5.2 Registration Module (Parent & NGO Accounts Creation)
    - 5.3 Profile Management Module
    - 5.4 M-CHAT-R Screening Module
    - 5.5 Notification / Reminder Module
    - 5.6 Reporting & Analytics Module
    - 5.7 Add-On Module
        - 5.7.1 Behaviour Monitoring

- 5.8 Integrate all modules into the system

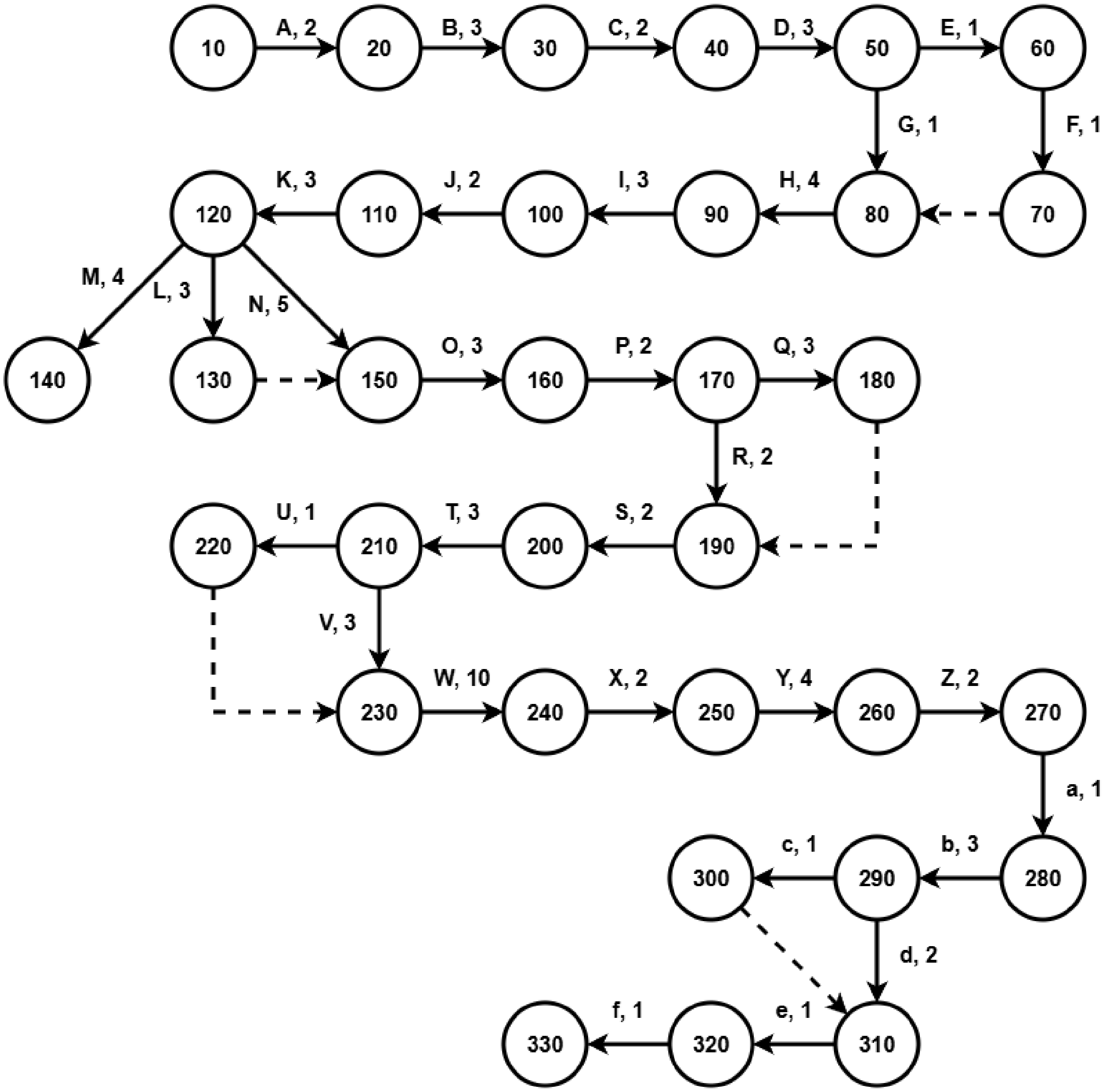
6. Testing
    - 6.1 Unit Testing for Each Module
    - 6.2 System Testing
    - 6.3 Bug Fixing & Debugging
    - 6.4 User Acceptance Testing (UAT)
7. Deployment & Documentation
    - 7.1 System Deployment
    - 7.2 Preparing User Manual
    - 7.3 Final Report Submission
    - 7.4 Final Presentation



7.3 PERT Chart (based on WBS)

Activity ID	Activity	Predecessor	Duration (Days)
A	Identify Problem	None	2
B	Background Study	A	3
C	Define Scope	B	2
D	Conduct Feasibility Study	C	3
E	Resource Allocation	D	1
F	Set Timelines & Milestones	E	1
G	Perform Cost-Benefit Analysis (CBA)	D	1
H	Gather Requirements	F, G	4
I	Conduct User Research	H	3
J	Analyse M-CHAT-R Screening	I	2
K	Develop Entity Relationship Diagram (ERD)	J	3
L	Design System Architecture	K	3
M	Design UI / UX Wireframes	K	4
N	Database Design	K	5
O	API & Integration Planning	L, N	3
P	Security Framework	O	2
Q	Design Authentication Module	P	3
R	Design Registration Module	P	2
S	Design Profile Management Module	Q, R	2
T	Design M-CHAT-R Screening Module	S	3
U	Design Notification / Reminder Module	T	1

V	Design Reporting & Analytics Module	T	3
W	Design Behaviour Monitoring Module	U, V	10
X	Integrate System Modules	Q, R, S, T, U, V, W	2
Y	Conduct Unit Testing for Each Module	X	4
Z	Conduct System Testing	Y	2
a	Conduct Bug Fixing & Debugging	Z	1
b	Conduct User Acceptance Testing (UAT)	a	3
c	System Deployment	b	1
d	Prepare User Manual	b	2
e	Final Report Submission	c, d	1
f	Final Presentation	e	1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Z, 2
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c, 1
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b, 3
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d, 2
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f, 1
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e, 1
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7.4 Gantt Chart

Activity ID	Activity Name	Duration	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
A	Identify Problem	2 days											
B	Background Study	3 days											
C	Define Scope	2 days											
D	Conduct Feasibility Study	3 days											
E	Resource Allocation	1 day											
F	Set Timelines & Milestones	1 day											
G	Perform Cost-Benefit Analysis (CBA)	1 day											
H	Gather Requirements	4 days											
I	Conduct User Research	3 days											
J	Analyse M-CHAT-R Screening	2 days											
K	Develop Entity Relationship Diagram (ERD)	3 days											
L	Design System Architecture	3 days											
M	Design UI / UX Wireframes	4 days											
N	Database Design	5 days											
O	API & Integration Planning	3 days											
P	Security Framework	2 days											
Q	Design Authentication Module	3 days											
R	Design Registration Module	2 days											
S	Design Profile Management Module	2 days											
T	Design M-CHAT-R Screening Module	3 days											
U	Design Notification / Reminder Module	1 day											
V	Design Reporting & Analytics Module	3 days											
W	Design Behaviour Monitoring Module	10 days											
X	Integrate System Modules	2 days											
Y	Conduct Unit Testing for Each Module	4 days											
Z	Conduct System Testing	2 days											
a	Conduct Bug Fixing & Debugging	1 day											
b	Conduct User Acceptance Testing (UAT)	3 days											
c	System Deployment	1 day											
d	Prepare User Manual	2 days											
e	Final Report Submission	1 day											
f	Final Presentation	1 day											

8.0 BENEFIT AND OVERALL SUMMARY OF PROPOSED SYSTEM

The Autism Early Screening Support System represents a comprehensive solution designed to transform early autism intervention for children. Unlike traditional screening approaches that focus solely on detection, this system integrates three components which is early screening along with additional modules including daily routine management, and behavior monitoring. This integrated approach provides continuous support for children with Autism Spectrum Disorder (ASD) and their families throughout the intervention journey.

The system's benefits are multifaceted:

1. **Enhanced Early Detection:**
   The M-CHAT-R significantly improves early detection because of its design with very high sensitivity in an attempt to catch as many potential autism cases as possible. The tool can be administered during regular checkup appointments with a pediatrician or healthcare provider for healthy children at 1 to 4 years old, turning such appointments into screening opportunities. This systematic approach catches children who might otherwise go unnoticed until much later.

   Early detection matters because autism interventions are most effective when started during critical developmental periods in toddlerhood. The earlier autism is recognized, the sooner children can receive targeted support, which significantly improves their long-term outcomes.

2. **Holistic Support for Families:**
   The Daily Routine Management component provides visual schedules and structured routines that are essential for children with ASD, who often thrive on predictability and clear expectations. Visual supports such as picture schedules showing morning routines or transition activities help children understand what comes next, reducing anxiety and improving independence.

   The Behavior Monitoring feature tracks patterns over time, such as when meltdowns occur, what triggers them, or which calming strategies work best. Parents gain actionable insights rather than just relying on memory, helping them identify environmental factors, dietary influences, or sleep patterns that affect their child's behavior. This data-driven approach transforms parenting from reactive to proactive.

3. **Improved Clinical Decision-Making:**
   Healthcare professionals typically see children for brief appointments, making it challenging to understand day-to-day realities. Comprehensive reporting that combines behavioral logs, routine adherence data, and parent observations gives clinicians a fuller picture of each child's functioning across different settings and times.

This enriched data enables personalized intervention strategies. For example, if the data shows a child struggles most during transitions, therapists can focus on transition-specific skills. Analytics can reveal progress trends, helping professionals adjust therapy intensity or focus areas based on real-world evidence rather than clinic-based observations alone.

4. **Increased Accessibility and Equity:**
   Mobile and web platforms remove geographical barriers by bringing screening tools directly to families' devices, whether they live in urban centers or rural areas. Families who face transportation challenges, work inflexibility, or live far from specialized clinics can now access screening and support resources without traveling.

   This is particularly impactful for underserved communities where early autism detection rates are traditionally lower due to limited access to pediatric specialists or awareness of screening tools.

5. **Efficient Resource Utilization:**
   NGOs and healthcare providers often struggle with paper-based systems, duplicate data entry, and fragmented information across multiple platforms. Digital integration streamlines administrative tasks, allowing screening results to automatically flow into databases, reports generate automatically, and case management becomes centralized.

   Better data collection enables organizations to track outcomes, identify service gaps, and demonstrate impact to funders. Improved care coordination means fewer missed appointments, better follow-up compliance, and clearer communication between parents, therapists, and medical providers, maximizing the effectiveness of limited resources.

In summary, this system is not merely a screening tool but a comprehensive support ecosystem designed to empower families, inform professionals, and ultimately improve the developmental trajectories and quality of life for children with autism spectrum disorder. It represents a significant step forward in leveraging technology for social good, aligning with the principles of accessible and data-driven healthcare.

9.0 REFERENCES

CDC. "Signs and Symptoms of Autism Spectrum Disorder." *CDC*, 16 May 2024, https://www.cdc.gov/autism/signs-symptoms/index.html. Accessed 3 June 2026.